

# Micro:bit Reaction Game Lesson Plan

## Lesson Overview

**Title:** Micro:bit Reaction Game - 2 Players **Grade Level:** 5th Grade **Subject Areas:** STEAM (Science, Technology, Engineering, Arts, Mathematics) - Focus on Computer Science, Engineering Design, and Physical Science **Duration:** 90-120 minutes (can be split into 2 sessions: building and coding)

**Description:** Students will design and build a physical game board using simple materials to create touch-sensitive switches, then program a Micro:bit to create a two-player reaction time game. This hands-on project introduces basic electronics, conductivity, programming concepts like variables, events, loops, conditionals, and randomness. Students will test reaction times, analyze data, and iterate on their designs.

## Learning Objectives

By the end of the lesson, students will be able to:

- Explain how simple circuits work using conductivity (e.g., human body completing a circuit via foil pads).
- Build a physical input device (game board) with constraints on materials.
- Program a Micro:bit using block-based coding (MakeCode) to handle events, timing, and logic for a reaction game.
- Test and debug their code and hardware, measuring reaction times in milliseconds.
- Analyze data from multiple trials to identify patterns (e.g., average reaction time).

## Aligned Standards

### Arizona Computer Science Standards (Grades 3-5)

- **3-5.CS.02:** Use appropriate tools and techniques to develop and debug programs.
- **3-5.AP.10:** Use algorithms to solve problems.
- **3-5.AP.13:** Decompose problems into smaller parts (e.g., building board, coding logic).
- **3-5.AP.14:** Create programs that include sequences, events, loops, and conditionals.
- **3-5.IC.19:** Use technology responsibly and ethically (e.g., safe handling of hardware).

### Arizona Science Standards (5th Grade)

- **5.P3U1.4:** Define a simple design problem reflecting a need or want that includes specified criteria for success and constraints on materials, time, or cost.
- **3-5.SEP.1:** Ask questions and define problems (e.g., "How can we fairly measure reaction time?").
- **3-5.SEP.3:** Plan and carry out investigations (e.g., testing the game).
- **3-5.SEP.4:** Analyze and interpret data (e.g., reaction times).

- **3-5.SEP.6:** Construct explanations and design solutions (e.g., explaining circuit conductivity).
- **3-5.CC.2:** Cause and effect (e.g., pressing pad completes circuit → triggers event).

## Common Core Math Standards (5th Grade)

- **5.MD.B.2:** Represent and interpret data (e.g., line plots of reaction times).

## NGSS Connections (if applicable, based on similar lessons)

- **4-PS3-2:** Make observations to provide evidence that energy can be transferred from place to place by electric currents.
- **3-5-ETS1-1:** Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints on materials, time, or cost.

## Materials (Per Group of 2-4 Students)

- 1 BBC Micro:bit board
- 1 Micro:bit battery holder with 2 AAA batteries
- 4 crocodile clip leads (alligator clips)
- Scrap cardboard (e.g., cereal box)
- Aluminum foil
- Glue, scissors, permanent markers
- Computer or tablet with access to MakeCode ([makecode.microbit.org](https://makecode.microbit.org))
- USB cable for downloading code to Micro:bit
- Optional: Tape, colored paper for decorating the board

## Preparation

- Set up computers with MakeCode editor open.
- Pre-watch introductory videos (e.g., "What is a Micro:bit?" if available).
- Group students in pairs for the 2-player game.
- Ensure safety: Remind students to handle electronics gently and avoid short-circuiting pins.

## Lesson Procedures

### Introduction (10-15 minutes)

1. **Hook:** Ask students, "What's your reaction time? How fast can you respond to a signal?" Demonstrate a simple reaction test (e.g., drop a ruler and catch it). Discuss real-world applications like sports, gaming, or driving.
2. **Introduce Micro:bit:** Show the device and explain its inputs (pins for touch), outputs (LED display), and programming via MakeCode. Play a short video on what a Micro:bit is (e.g., from [microbit.org](https://microbit.org)).
3. **Overview the Project:** Explain that students will build a game board with foil pads as switches, code the Micro:bit to wait randomly, show a signal, detect who reacts first, and display the

winner or time. Connect to concepts: conductivity (body completes circuit), randomness (fairness), and timing (variables).

4. **Review Objectives and Standards:** Share what they'll learn and why it matters.

### Activity 1: Building the Game Board (20-30 minutes)

1. **Design Problem:** Students define the problem - create two touch pads that act as buttons for players A and B, connected to Micro:bit pins. Criteria: Pads must be large enough to slap, durable, and conductive. Constraints: Use only provided materials.
2. **Materials and Steps:**
  - Cut cardboard into a base (e.g., 8x10 inches).
  - Create two pads: Glue foil squares (4x4 inches) to cardboard, leaving space between them.
  - For each pad, attach one foil side to GND (shared ground) and the other to Pin 1 (Player A) or Pin 2 (Player B) using crocodile clips. (Note: Players touch both foils to complete the circuit via their body.)
  - Decorate with markers (e.g., label "Player A" and "Player B").
  - Test conductivity: Connect to Micro:bit and use a simple touch test in MakeCode.
3. **Engineering Practices:** Encourage sketching designs, testing prototypes, and iterating if pads don't work (e.g., add more foil for better contact).

### Activity 2: Coding the Game (30-45 minutes)

1. **Decompose the Problem:** Break into parts - countdown, random wait, signal, detect press, display winner/time.
2. **Step-by-Step Coding in MakeCode (Block-Based):**
  - **Variables:** Create start, end, running (boolean), false\_start (boolean). Initialize to 0 or false.
  - **Event: On Pin P0 Pressed (Start Button):** Use GND + P0 as a start pad if needed, or button A. Show countdown (3, 2, 1), clear screen, pause randomly (1000-3000 ms). If no false start, set start to running time, set running true, show signal (e.g., LED plot or heart).
  - **Event: On Pin P1 Pressed (Player A):** If running true, set running false, set end to running time, show "A" or left-side LEDs, display time (end - start). Else, false start: Show "X".
  - **Event: On Pin P2 Pressed (Player B):** Similar to P1, but show "B" or right-side LEDs.
  - **Logic for Winner:** Use a winner variable to track who won first and prevent the loser from triggering false start.
  - **Loop:** Wrap in a forever loop for multiple rounds.
3. **Debugging:** Download code to Micro:bit via USB, test with board. Fix issues like timing or false triggers.
4. **Alternative Simple Version (Python-Style Logic):** For advanced students, adapt the microbit.org code: Random sleep, show heart, check if Pin1 or Pin2 touched, show A or B.

### Activity 3: Testing, Data Analysis, and Play (15-20 minutes)

1. **Investigate:** Play 5-10 rounds, record reaction times in a table (e.g., Player A vs. B).
2. **Analyze:** Calculate averages, discuss patterns (e.g., "Does practice improve reaction time?"). Create a simple graph.
3. **Share:** Groups demo their games and explain how the circuit/code works.

### Assessment

- **Formative:** Observe participation in building/coding, check for working game board and code.
- **Summative:** Rubric on objectives (e.g., 4 points: Circuit works, code detects winner/time accurately; data analyzed).
- **Self-Reflection:** Students journal: "What was challenging? How did you debug?"
- **Exit Ticket:** Explain cause-effect in the circuit (e.g., "Touching pad causes...").

### Differentiation and Extensions

- **Support:** Provide pre-cut materials or starter code templates. Pair stronger coders with builders.
- **Challenge:** Add scores (variables to track wins), fastest time tracker, or multiplayer with more pins. Integrate radio for wireless multi-Micro:bit games. Convert to buzzwire game (continuous touch detection).
- **Cross-Curricular:** Art - Design themed boards; Math - Statistical analysis of times; Science - Experiment with different materials for conductivity.
- **Home Extension:** Students build variations at home and share videos.

This lesson fosters collaboration, problem-solving, and excitement for coding/physical computing while aligning to standards. For resources, visit [makecode.microbit.org/projects/reaction-time](https://makecode.microbit.org/projects/reaction-time) or [microbit.org/projects/make-it-code-it/reaction-game](https://microbit.org/projects/make-it-code-it/reaction-game). If needed, adapt based on class size or tech access!